

Estimation (Part 1)

STAT 4530/6530
Statistical Inference for Data Scientists

Statistical Inference: Estimation

A key part of statistical inference is estimating population parameters.

We can use sample data to estimate parameters in multiple ways:

- A **point estimate** is the value of a statistic that predicts the parameter value. An **estimator** is a particular statistic used to construct a point estimate. Estimators take on different values for different samples.
- A **interval estimate** is an interval of numbers around the point estimate within which the true parameter value is believed to fall with some degree of “confidence.” For now, we consider what is called a “confidence interval”
- We assume (for now) that each parameter of interest has an unknown but fixed true value. In other words, we do **not** assume that the parameter is a random variable that can take on a range of plausible values with varying probability.

Properties of Estimators

Consider an estimator $\hat{\theta}$ of a parameter θ . Remember, an estimator is a statistic (a function of random variables that represent hypothetical data) and thus has a sampling distribution.

A good estimator will have a sampling distribution that (1) centers around θ and (2) has a small standard error. There are a number of ways of specifying such properties:

- An estimator is **unbiased** if $E(\hat{\theta}) = \theta$
- An estimator is **consistent** if $P(|\hat{\theta} - \theta| \geq \epsilon) \rightarrow 0$ as $n \rightarrow \infty$, that is, $\hat{\theta}$ converges in probability to θ
- A good estimator will tend to fall “closer” to θ , on average, and this “closeness” can be summarized by the **mean squared error** (MSE):

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

Mean Squared Error

Let us define the **bias** of an estimator as

$$\text{bias}(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

Then it can be shown that the mean squared error of an estimator is the sum of its variance and squared bias:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2.$$

Although unbiasedness is a desirable property, it is not necessarily desirable if it comes at the cost of high variability. It may, on the other hand, be better to accept some bias if the estimator has relatively small variance. Thus the mean squared error captures the dynamic of this bias/variance tradeoff.

An estimator is **efficient** if it has minimum mean squared error, uniformly for the possible values of θ .

Mean Squared Error

Example: Consider the number of “successes” Y generated by a $\text{Bin}(n, p)$ distribution.

We consider two different estimators of p :

- $\hat{p} = Y/n$
- $\tilde{p} = (Y + c)/(n + 2c)$ for a nonnegative value c

What is the bias and variance of each estimator?

Properties of Estimators

Many widely-used estimators are biased but they are **asymptotically unbiased**, meaning that the degree of bias tends to 0 as the sample size n increases

As an example, consider estimating the population variance σ^2 from independent random variables $X_1, \dots, X_n$:

- The estimator

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is an unbiased estimator of σ^2

- The estimator

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

is biased, but it is asymptotically unbiased

Properties of Estimators

Just as we can consider asymptotic unbiasedness, we can define an estimator as being **asymptotically efficient** if its MSE divided by an efficient estimator converges to 1 as the sample size n increases

In summary, a good estimator is consistent and (at least) asymptotically unbiased and asymptotically efficient

- For estimating a population proportion with independent observations (the proportion of “successes” in independent Bernoulli trials), the sample proportion is consistent, unbiased and efficient
- For estimating a population mean μ , the sample mean $\bar{X}$ is consistent (by the law of large numbers), it is unbiased, and it is efficient for many population distributions (such as the normal distribution)
- However, the sample median is not an efficient or asymptotically efficient estimator of the median of most distributions

Interval Estimation

To be truly informative, we want to know how close an estimator $\hat{\theta}$ will likely be to the parameter θ that it estimates.

The precision of $\hat{\theta}$ determines the width of an interval estimate of θ . Because we cannot be sure that an interval estimate contains θ but merely that it does so with a certain degree of confidence, it is called a confidence interval.

A **confidence interval** for a parameter θ is an interval of numbers within which θ is predicted to fall.

- The method for constructing confidence intervals depends on the sample of data, and a different confidence interval will be constructed for any given sample.
- The probability that the method for constructing confidence intervals produces an interval that truly contains θ is called the **confidence level**.
- The confidence level is typically chosen to be a number close to 1, such as 0.95 or 0.99.

Interval Estimation

One way to construct a confidence interval uses the sampling distribution of the estimator $\hat{\theta}$. Consider $\hat{\theta} = \bar{X} = \sum_{i=1}^n X_i$, where $X_1, \dots, X_n$ are independent random variables from a population with mean θ .

- The sampling distribution in this case is asymptotically normal with mean θ
- The normal distribution and its standard error then determines the probability that $\hat{\theta}$ falls within a certain distance of θ
- To construct a confidence interval, we can add and subtract from $\hat{\theta}$ some multiple of the standard error.
- For example, with probability of approximately 0.95, $\hat{\theta}$ falls within two standard errors of θ , so to have “95% confidence,” we take the point estimate given by $\hat{\theta}$ for a particular sample and add and subtract a margin of error that equals about two standard errors.

Maximum Likelihood Estimation

Before analyzing data, we typically assume that the data have been generated by a particular parametric family of probability distributions.

- Different families of distributions are appropriate depending on the nature of the data. For instance, are they discrete or continuously distributed? Are the values restricted to lie within certain ranges? Are they binary?

Under the assumption that a particular parametric family of distributions generated the data, the task is then to fit the distribution to the data by estimating the parameter or parameters that characterize the distribution.

- We will develop a way of constructing estimators that are consistent, asymptotically unbiased and asymptotically efficient: the point estimate is the parameter value for which the observed data would have been most likely to occur.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.